

```
! pip install kaggle
from google.colab import drive
drive.mount('/content/drive')
! mkdir ~/.kaggle
! cp kaggle.json ~/.kaggle
! chmod 600 ~/.kaggle/kaggle.json
! kaggle datasets download 'paultimothymooney/breast-histopathology-images'
! unzip breast-histopathology-images.zip
```

```
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1751_y951_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1801_y1001_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1801_y1051_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1801_y601_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1801_y651_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1801_y701_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1801_y751_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1801_y801_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1801_y851_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1801_y901_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1801_y951_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1851_y1001_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1851_y1051_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1851_y601_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1851_y651_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1851_y701_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1851_y751_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1851_y801_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1851_y851_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1851_y901_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1851_y951_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y1001_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y1051_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y601_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y651_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y701_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y751_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y801_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y851_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y901_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y951_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y1001_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y1051_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y601_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y651_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y701_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y751_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y801_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y851_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y901_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y951_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2001_y1001_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2001_y1051_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2001_y701_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2001_y751_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2001_y801_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2001_y851_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2001_y901_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2051_y1051_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2051_y801_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2051_y851_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2051_y901_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2051_y951_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2101_y1001_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2101_y901_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2101_y951_class1.png
```

```
Automatic saving failed. This file was updated remotely or in another tab. Show diff
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y1001_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y1051_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y601_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y651_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y701_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y751_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y801_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y851_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y901_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1901_y951_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y1001_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y1051_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y601_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y651_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y701_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y751_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y801_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y851_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y901_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x1951_y951_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2001_y1001_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2001_y1051_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2001_y701_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2001_y751_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2001_y801_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2001_y851_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2001_y901_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2051_y1051_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2051_y801_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2051_y851_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2051_y901_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2051_y951_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2101_y1001_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2101_y901_class1.png
inflating: IDC_regular_ps50_idx5/9383/1/9383_idx5_x2101_y951_class1.png
```

```
[2] import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import cv2
import glob
import random
from sklearn.metrics import classification_report
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense, Dropout, BatchNormalization
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping
from tensorflow.keras.utils import to_categorical
```

```
Automatic saving failed. This file was updated remotely or in another tab. Show diff
from tensorflow.keras.applications import VGG19
from tensorflow.keras.layers import Input
```

```
[3] from google.colab import drive
drive.mount('/content/drive')

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
[4] breast_img = glob.glob('/content/IDC_regular_ps50_idx5/**/*.png', recursive = True)

for imgname in breast_img[:3]:
    print(imgname)

/content/IDC_regular_ps50_idx5/9174/1/9174_idx5_x1551_y901_class1.png
```

```
/content/IDC_regular_ps50_idx5/9174/1/9174_idx5_x1401_y1101_class1.png
/content/IDC_regular_ps50_idx5/9174/1/9174_idx5_x1501_y801_class1.png
```

```
0s [5] # Split images based on class (IDC negative and IDC positive)
N_IDC = [img for img in breast_img if img[-5] == '0']
P_IDC = [img for img in breast_img if img[-5] == '1']
```

Automatic saving failed. This file was updated remotely or in another tab. [Show diff](#)

```
0s [7] # Data Preprocessing
non_img_arr = []
can_img_arr = []
```

```
23s [8] # Process IDC (-) images
for img in NewN_IDC:
    n_img = cv2.imread(img, cv2.IMREAD_COLOR)
    n_img_size = cv2.resize(n_img, (50, 50), interpolation=cv2.INTER_LINEAR)
    non_img_arr.append([n_img_size, 0])
```

```
21s [9] # Process IDC (+) images
for img in P_IDC:
    c_img = cv2.imread(img, cv2.IMREAD_COLOR)
    c_img_size = cv2.resize(c_img, (50, 50), interpolation=cv2.INTER_LINEAR)
    can_img_arr.append([c_img_size, 1])
```

```
0s [10] # Combine the processed images and shuffle
breast_img_arr = np.concatenate((non_img_arr[:12389], can_img_arr[:12389]))
random.shuffle(breast_img_arr)

X = np.array([feature for feature, _ in breast_img_arr])
```

Automatic saving failed. This file was updated remotely or in another tab. [Show diff](#)

```
<_array_function__ internals>:180: VisibleDeprecationWarning: Creating an ndarray from ragged nested sequences (which is a list-or-tuple of lists-or-tuples-or ndarrays with different l
```

4

```
0s [11] # Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3)
```

```
1s [12] # Normalize pixel values
X_train = X_train / 255.0
X_test = X_test / 255.0
```

```
2s [13] vgg_base = VGG19(weights='imagenet', include_top=False, input_tensor=Input(shape=(50, 50, 3)))
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/vgg19/vgg19_weights_tf_dim_ordering_tf_kernels_notop.h5
80134624/80134624 [=====] - 1s 0us/step
```

```
0s [14] for layer in vgg_base.layers:
    layer.trainable = False
```

```
0s [15] from tensorflow.keras.models import Model
```

Automatic saving failed. This file was updated remotely or in another tab. [Show diff](#)

```
x = Flatten()(x)
x = Dense(128, activation='relu', kernel_initializer='he_uniform')(x)
x = BatchNormalization()(x)
x = Dropout(0.5)(x)
predictions = Dense(1, activation='sigmoid')(x)

model_vgg = Model(inputs=vgg_base.input, outputs=predictions)
```

```
0s [16] from tensorflow.keras.optimizers import Adam

model_vgg.compile(optimizer=Adam(learning_rate=0.0001), loss='binary_crossentropy', metrics=['accuracy'])
```

```
0s [17] from tensorflow.keras.preprocessing.image import ImageDataGenerator
```

```
train_datagen_vgg = ImageDataGenerator(
    rotation_range=20,
    width_shift_range=0.2,
    height_shift_range=0.2,
    shear_range=0.2,
    zoom_range=0.2,
    horizontal_flip=True,
    vertical_flip=True)
```

Automatic saving failed. This file was updated remotely or in another tab. [Show diff](#)

```
train_generator_vgg = train_datagen_vgg.flow(X_train, y_train, batch_size=35)
```

```
1h [18] history_vgg = model_vgg.fit(train_generator_vgg, steps_per_epoch=len(X_train) // 35,
    validation_data=(X_test, y_test), epochs=200)
```

```
Epoch 172/200
495/495 [=====] - 26s 52ms/step - loss: 0.2432 - accuracy: 0.9046 - val_loss: 0.2625 - val_accuracy: 0.9010
Epoch 173/200
495/495 [=====] - 26s 53ms/step - loss: 0.2487 - accuracy: 0.9030 - val_loss: 0.2563 - val_accuracy: 0.9019
Epoch 174/200
495/495 [=====] - 28s 57ms/step - loss: 0.2437 - accuracy: 0.9025 - val_loss: 0.2714 - val_accuracy: 0.8939
Epoch 175/200
495/495 [=====] - 25s 50ms/step - loss: 0.2411 - accuracy: 0.9055 - val_loss: 0.2562 - val_accuracy: 0.9033
Epoch 176/200
```

```
495/495 [=====] - 25s 51ms/step - loss: 0.2464 - accuracy: 0.9024 - val_loss: 0.2533 - val_accuracy: 0.9052
Epoch 177/200
495/495 [=====] - 27s 54ms/step - loss: 0.2397 - accuracy: 0.9073 - val_loss: 0.2586 - val_accuracy: 0.8997
Epoch 178/200
495/495 [=====] - 25s 50ms/step - loss: 0.2422 - accuracy: 0.9051 - val_loss: 0.2590 - val_accuracy: 0.9023
Epoch 179/200
495/495 [=====] - 28s 56ms/step - loss: 0.2422 - accuracy: 0.9057 - val_loss: 0.2650 - val_accuracy: 0.9002
Epoch 180/200
495/495 [=====] - 3s 2444 - accuracy: 0.9037 - val_loss: 0.2538 - val_accuracy: 0.9066
```

Automatic saving failed. This file was updated remotely or in another tab. [Show diff](#)

```
495/495 [=====] - 26s 53ms/step - loss: 0.2420 - accuracy: 0.9051 - val_loss: 0.2683 - val_accuracy: 0.8953
Epoch 182/200
495/495 [=====] - 28s 56ms/step - loss: 0.2448 - accuracy: 0.9034 - val_loss: 0.2481 - val_accuracy: 0.9070
Epoch 183/200
495/495 [=====] - 26s 53ms/step - loss: 0.2410 - accuracy: 0.9053 - val_loss: 0.2630 - val_accuracy: 0.8957
Epoch 184/200
495/495 [=====] - 26s 53ms/step - loss: 0.2442 - accuracy: 0.9017 - val_loss: 0.2556 - val_accuracy: 0.9036
Epoch 185/200
495/495 [=====] - 29s 58ms/step - loss: 0.2420 - accuracy: 0.9052 - val_loss: 0.2571 - val_accuracy: 0.9010
Epoch 186/200
495/495 [=====] - 26s 53ms/step - loss: 0.2418 - accuracy: 0.9042 - val_loss: 0.2619 - val_accuracy: 0.9018
Epoch 187/200
495/495 [=====] - 25s 50ms/step - loss: 0.2438 - accuracy: 0.9030 - val_loss: 0.2619 - val_accuracy: 0.9045
Epoch 188/200
495/495 [=====] - 28s 57ms/step - loss: 0.2408 - accuracy: 0.9056 - val_loss: 0.2690 - val_accuracy: 0.8953
Epoch 189/200
495/495 [=====] - 26s 53ms/step - loss: 0.2434 - accuracy: 0.9040 - val_loss: 0.2673 - val_accuracy: 0.8964
Epoch 190/200
495/495 [=====] - 25s 51ms/step - loss: 0.2394 - accuracy: 0.9063 - val_loss: 0.2640 - val_accuracy: 0.8951
Epoch 191/200
495/495 [=====] - 25s 51ms/step - loss: 0.2437 - accuracy: 0.9031 - val_loss: 0.2607 - val_accuracy: 0.8980
Epoch 192/200
495/495 [=====] - 25s 50ms/step - loss: 0.2440 - accuracy: 0.9023 - val_loss: 0.2619 - val_accuracy: 0.8990
Epoch 193/200
495/495 [=====] - 27s 54ms/step - loss: 0.2399 - accuracy: 0.9061 - val_loss: 0.2532 - val_accuracy: 0.9015
Epoch 194/200
495/495 [=====] - 25s 50ms/step - loss: 0.2386 - accuracy: 0.9044 - val_loss: 0.2608 - val_accuracy: 0.8956
```

Automatic saving failed. This file was updated remotely or in another tab. [Show diff](#)

```
495/495 [=====] - 3s 2405 - accuracy: 0.9046 - val_loss: 0.2425 - val_accuracy: 0.9073
Epoch 196/200
495/495 [=====] - 25s 51ms/step - loss: 0.2384 - accuracy: 0.9059 - val_loss: 0.2567 - val_accuracy: 0.9006
Epoch 197/200
495/495 [=====] - 27s 54ms/step - loss: 0.2422 - accuracy: 0.9051 - val_loss: 0.2758 - val_accuracy: 0.8894
Epoch 198/200
495/495 [=====] - 26s 53ms/step - loss: 0.2421 - accuracy: 0.9042 - val_loss: 0.2475 - val_accuracy: 0.9088
Epoch 199/200
495/495 [=====] - 26s 53ms/step - loss: 0.2441 - accuracy: 0.9047 - val_loss: 0.2573 - val_accuracy: 0.9015
Epoch 200/200
495/495 [=====] - 27s 54ms/step - loss: 0.2430 - accuracy: 0.9049 - val_loss: 0.2604 - val_accuracy: 0.8997
```

✓ [19] model_vgg.evaluate(X_test, y_test)

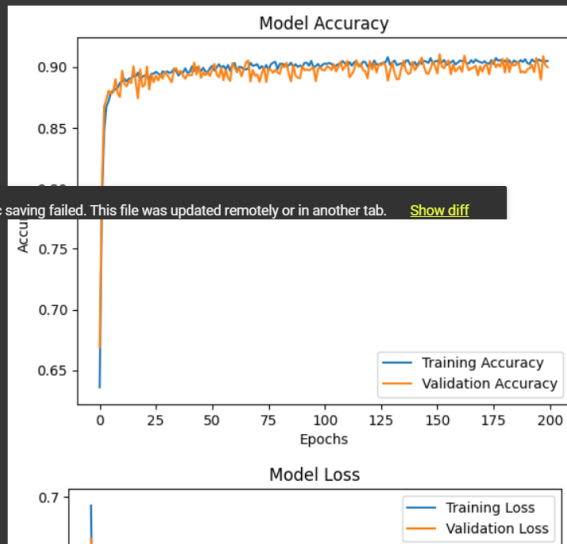
```
233/233 [=====] - 4s 17ms/step - loss: 0.2604 - accuracy: 0.8997
[0.26037126779556274, 0.8996502757072449]
```

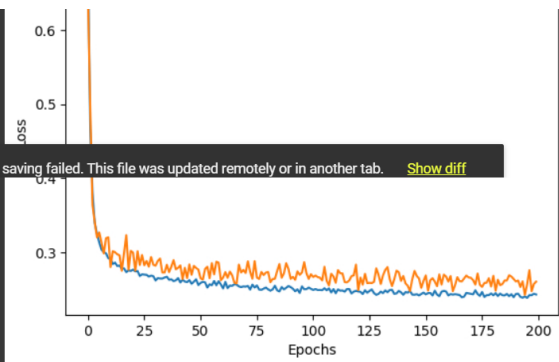
✓ [20] model_vgg.save("Brest_idc_VGG19_Model.h5")

```
/usr/local/lib/python3.10/dist-packages/keras/src/engine/training.py:3103: UserWarning: You are saving your model as an HDF5 file via `model.save()`. This file format is considered legacy. We recommend using the newer Keras format via `model.save_format='keras'` instead.
saving_api.save_model(
```

```
Automatic saving failed. This file was updated remotely or in another tab. Show diff
plt.plot(history_vgg.history['val_accuracy'], label='Validation Accuracy')
plt.title('Model Accuracy')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()
plt.show()

plt.plot(history_vgg.history['loss'], label='Training Loss')
plt.plot(history_vgg.history['val_loss'], label='Validation Loss')
plt.title('Model Loss')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.show()
```





Automatic saving failed. This file was updated remotely or in another tab. [Show diff](#)

```
[24] # Predict probabilities for the test set
y_pred_prob = model_vgg.predict(X_test)

# Convert probabilities to classes
y_pred = (y_pred_prob > 0.5).astype('int32')

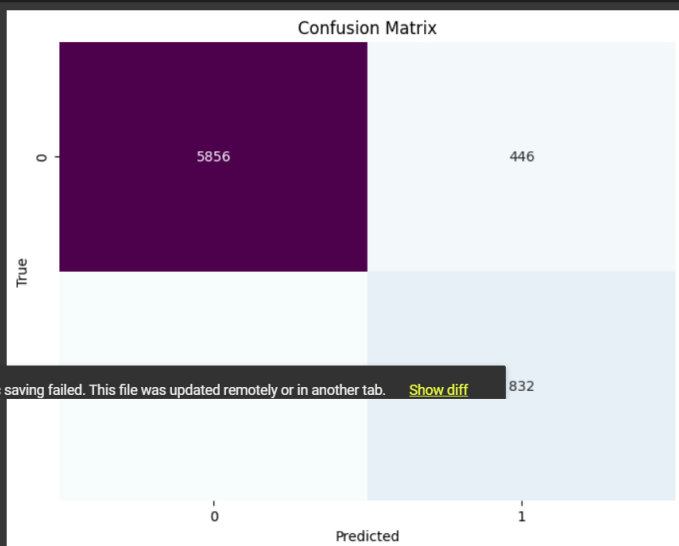
# Create confusion matrix
cm = confusion_matrix(y_test, y_pred)
```

233/233 [=====] - 4s 15ms/step

```
[25] # Plot Confusion Matrix
```

Automatic saving failed. This file was updated remotely or in another tab. [Show diff](#)

```
sns.heatmap(cm, annot=True, fmt='d', cmap='BuPu', cbar=False)
plt.xlabel('Predicted')
plt.ylabel('True')
plt.title('Confusion Matrix')
plt.show()
```



Automatic saving failed. This file was updated remotely or in another tab. [Show diff](#)

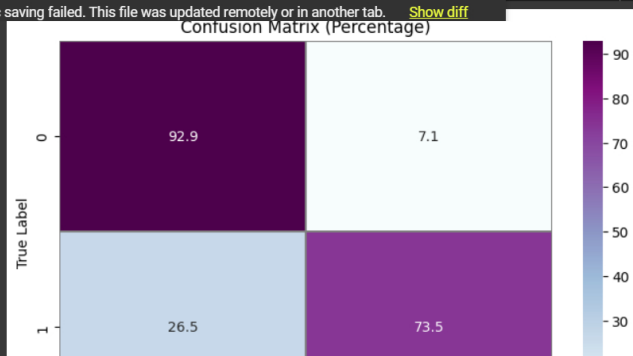
```
from sklearn.metrics import accuracy_score, confusion_matrix

confusion_mtx = confusion_matrix(y_test, y_pred)

# calculate the percentage
confusion_mtx_percent = confusion_mtx.astype('float') / confusion_mtx.sum(axis=1)[:, np.newaxis] * 100

f, ax = plt.subplots(figsize=(8, 5))
sns.heatmap(confusion_mtx_percent, annot=True, linewidths=0.01, cmap="BuPu", linecolor="gray", fmt='.1f', ax=ax)
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.title("Confusion Matrix (Percentage)")
plt.show()
```

Automatic saving failed. This file was updated remotely or in another tab. [Show diff](#)





[] Start coding or [generate](#) with AI.

Automatic saving failed. This file was updated remotely or in another tab. [Show diff](#)

[Colab paid products](#) - [Cancel contracts here](#)

✓ 0s completed at 3:52 PM

